Machine Learning Mini Project
UE24CS352A

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TITLE :- DETECTING AND PREDICTING EARTHQUAKE GROUNDMOTION DIRECTIONALITY

1. Problem Statement & Motivation

Earthquake ground motion is not necessarily strongest in the same horizontal direction at every recording site. The orientation associated with maximum horizontal response can vary from one site to another, even during the same earthquake. Although engineering practice often treats this orientation as uniformly random, observations indicate that nearby sites can exhibit related directionality patterns. Understanding this spatial relationship is important for characterising ground-motion variability and for assessing whether directionality at one site can provide information about another nearby site.

The study formulates the task as a supervised regression problem. For two recording sites affected by the same earthquake, the objective is to predict the absolute angular difference between their orientations of maximum ground-motion response, using the distance between sites and relevant earthquake, source, path and site attributes.

2. Dataset and Pair Construction

The study uses the PEER NGA-West2 database, which contains ground-motion records and associated earthquake and site metadata. Records are filtered to shallow-crustal earthquakes with magnitude M ‡ 5.0. The database flatfile provides a broad set of candidate attributes, while waveform records are used to calculate the directionality target, which is not directly supplied as a flatfile field.

The analysis constructs pairs of distinct recording sites from the same earthquake and applies a strict inter-site separation criterion of 5 km. This restriction focuses the study on nearby sites, where spatial correlation in directionality is most relevant. The paper reports approximately 5,300 eligible site pairs from a much larger set of candidate pairs. Candidate predictors are reduced through feature screening and selection before the final model comparison.

3. Estimating Orientation and Defining the Target

For each record, the two horizontal acceleration components are used to evaluate the response over horizontal orientations. A damped single-degree-of-freedom oscillator provides the response measure. For an orientation q, the horizontal motion is represented as a rotated combination of the two components; the response is evaluated over orientations spanning 0° to 179°. The orientation with the maximum response is taken as the directionality orientation for that record.

Because a horizontal orientation is unchanged by a 180° reversal, the angular difference is treated as a 180°-periodic quantity and folded into the interval [0°, 90°]. For sites i and j, the target is the absolute difference between their maximum-response orientations. This target allows the study to model whether the directionality at one site is related to that at a nearby site during the same earthquake.

4. Predictive Models and Evaluation

• Distance-only benchmark: a simple linear regression that predicts angular difference using inter-site distance alone. It establishes how much predictive information is available from separation distance without additional attributes.

• Ridge regression: a linear model with an L2 regularisation penalty. Feature selection and regularisation tuning are used to identify a compact predictor set while controlling model complexity.

• Neural network: a feed-forward network trained to model nonlinear relationships between the pair-level predictors and angular difference. Network configuration and regularisation are selected using the study's model-selection procedure.

The models are assessed on held-out data using the coefficient of determination (R^2) and mean absolute error (MAE), with lower MAE indicating smaller average angular prediction errors. Comparing a simple benchmark, a regularised linear model and a neural network helps determine whether additional features and nonlinear modelling improve prediction beyond distance alone.

5. Results

The paper reports the following held-out evaluation results for the distance benchmark, Ridge regression and neural network:

Model	R^2	MAE
Distance-only benchmark	0.025	20.2°
Ridge regression	0.245	16.9°
Neural network	0.354	13.2°

Among the reported models, the neural network achieves the highest R^2 and lowest MAE, indicating the strongest predictive performance of the three approaches. Ridge regression also improves on the distance-only benchmark. These results suggest that additional earthquake and site attributes, together with nonlinear relationships, can help explain variation in directionality differences between nearby sites.

6. Interpretation and Significance

The study's central contribution is a data-driven framework for predicting the difference in maximum-response orientations between nearby recording sites. The distance-only benchmark provides a baseline, while Ridge regression and the neural network test whether richer features and more flexible function approximations improve predictive accuracy. The reported improvement from the benchmark to the neural network supports the potential value of modelling directionality as a relationship among site, source and path characteristics rather than as a purely random quantity.

The work also illustrates the importance of the spatial scale used in the analysis. By concentrating on pairs within 5 km, the study focuses on a regime in which nearby sites may share aspects of the directionality pattern. The findings should therefore be interpreted in the context of the paper's pair-selection criteria, target definition, feature set and evaluation procedure.

7. Conclusion

The proposed research presents a supervised-learning approach to estimating ground-motion directionality differences between nearby sites affected by the same earthquake. It derives orientation labels from horizontal acceleration records, constructs same-event site pairs, and compares a distance-only benchmark with Ridge regression and a neural network. The reported results favour the neural network, followed by Ridge regression, over the distance-only benchmark. Overall, the paper demonstrates how waveform-derived directionality measures and earthquake/site metadata can be combined to investigate spatial patterns in strong ground motion.

Reference

Poulos, A. and Silva-Lopez, R., "Detecting and Predicting Earthquake Ground Motion Directionality Patterns Using Machine Learning Tools." Research paper supplied for the mini-project